

1 Important Constants & Equations

This week you got your first peak into circuit analysis in the context of physics. You were more properly introduced to capacitors along with series and parallel circuits components.

As we touched on last week, the electric potential and electric field are very closely related. Geometrically, the voltage is the area under the electric field curve:

$$\Delta V = V_f - V_i = - \int_{s_i}^{s_f} E_s ds = - \int_i^f \vec{E} \cdot d\vec{s} \quad (\text{The Voltage Difference Generated by an Electric Field})$$

Conversely, the electric field is the change in voltage per unit distance:

$$E = -\frac{dV}{ds} \implies E = E_x\hat{i} + E_y\hat{j} + E_z\hat{k} = -\left(\frac{\partial V}{\partial x}\hat{i} + \frac{\partial V}{\partial y}\hat{j} + \frac{\partial V}{\partial z}\hat{k}\right) \quad (\text{E-Field of a Potential Diff.})$$

One of the key laws you will use in circuit analysis is **Kirchhoff's Loop Law**. It essentially states that a closed loop forms a voltage difference of zero:

$$\Delta V_{\text{loop}} = \sum_i (\Delta V)_i = 0$$

The first circuit element (besides a battery) is a capacitor. Its key property is its ability to hold charge separated by a distance. This information is encoded in its capacitance value:

$$C = \frac{Q}{\Delta V_C} = \frac{\epsilon_0 A}{d} \implies Q_C = C \Delta V_C = \frac{\epsilon_0 A}{d} \Delta V_C \quad (\text{Definition of a Capacitor \& its Charge})$$

Lastly, the potential energy stored in a capacitor is:

$$U_C = \frac{1}{C} \int_0^Q q dq = \frac{Q^2}{2C} = \frac{1}{2} C (\Delta V_C)^2 = \frac{\epsilon_0}{2} (Ad) E^2 \quad (\text{Potential Energy of a Capacitor})$$

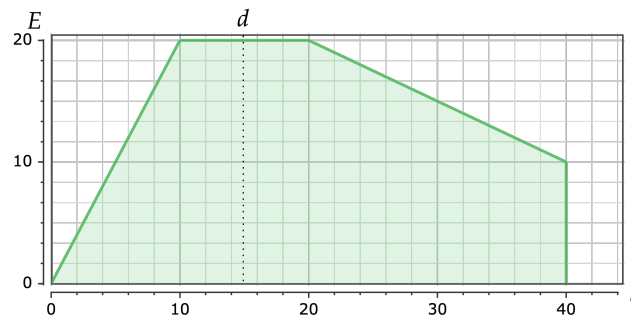
The following constants/relations are important for the study of electricity:

$$\begin{array}{lll} e = 1.60 \times 10^{-19} \text{ C} & (\text{Fundamental Unit of Charge}) & m_p = 1.67 \times 10^{-27} \text{ kg} \quad (\text{Mass of Proton}) \\ m_e = 9.11 \times 10^{-31} \text{ kg} & (\text{Mass of Electron}) & K = 8.99 \times 10^9 \text{ Nm}^2/\text{C}^2 \quad (\text{Electrostatic Constant}) \\ \epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/\text{Nm}^2 & (\text{Permittivity Constant}) & \lambda = dq/dl \quad \eta = dq/dA \quad \rho = dq/dV \end{array}$$

What to do with all of this empty space!

2 Problems

1. Below is a graph of an electric field E (in N/C) versus distance s (in meters).



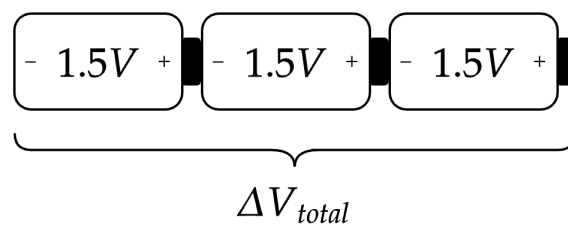
- (a) What is the electric field at the position d ?
- (b) What is the potential at the position d ?
2. A parallel plate capacitor of length s and plate area A is constructed in a way in which the magnitude of the electric field evolves with the function $E(x) = 5x^2$ N/C where x is the distance from left to right in the capacitor.
- (a) Draw a diagram of this situation. You do not have to indicate the increasing electric field.
- (b) What is the potential difference function $\Delta V(x)$ of this system?
- $\Delta V(x) =$

(c) Find an equation for the total charge Q stored in the capacitor.

(d) Find an equation for the capacitance C of the capacitor.

(e) Find an equation for the potential energy U_C stored in the capacitor.

3. As shown below, three AAA batteries are placed in series.



(a) What is the voltage across all three batteries?

$\Delta V_{total} =$

P.S. This is how a lot of replaceable-battery-powered devices work!

- (b) Now imagine you were to place a battery with $\varepsilon = \Delta V_{\text{total}}$ with three capacitors C_1 , C_2 , and C_3 in parallel. Draw an example of what the circuit may look like below.

- (c) Let $C_1 = 5\mu C$, $C_2 = 7\mu C$, and $C_3 = 10\mu C$. What would be the equivalent capacitance in this circuit?

- (d) Now let all of the capacitors have any random amount of capacitance. What would the voltage across all three capacitors be? Why?

4. **(CHALLENGE PROBLEM).** Your friend comes up with a supposed-ingenious idea for a capacitor: a *spherical capacitor*. While he hasn't worked out how you're actually supposed to charge it, you don't ask questions and he still requests you to find the capacitance per unit volume of the spherical capacitor.

- (a) The spherical capacitor consists of two spheres, each with a uniform charge density $\pm\rho$. The inner sphere has a radius R_1 with a negative charge and the outer sphere is hollow, positively-charged sphere with an inner radius of R_2 and outer radius of R_3 . The inner sphere and inner radius of the outer sphere are separated by a distance s .

Diagram the scenario. Indicate the different radii and the distance between the two spheres.

(b) Find an expression for the electric field E_s of the spherical capacitor.

(c) Find the voltage difference ΔV between the two spheres.

(d) Find the capacitance per unit volume of the spherical capacitor.

